KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering | SAP Data Analytics

Procure-to-Pay (P2P) Process

Full Purchasing Cycle in SAP S/4HANA Materials Management

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1. Problem Statement

Manual procurement lacks traceability, causing invoice-PO mismatches, overpayments, and delayed inventory updates. Without an integrated ERP, there is no automated control to confirm that what was ordered, received, and invoiced all agree — exposing organizations to fraud, duplicate payments, and audit failures.

Sunrise Traders Pvt. Ltd. needed a structured, auditable procurement cycle for CP-RM1 Steel — from demand capture to payment — with automated three-way matching, purchasing controls, real-time inventory updates, and automatic financial postings on a single platform.

2. Solution & Key Features

The SAP S/4HANA Procure-to-Pay process (MM module) enforces an organizational control hierarchy (Company Code → Plant → Storage Location → Purchasing Org → Purchasing Group) and automates all financial postings across the full procurement lifecycle.

- **Purchase Order (ME21N):** Legally binding supplier contract; triggers budget commitment in the system.
- **Goods Receipt (MIGO):** Real-time inventory update via Movement Type 101 to Storage Location 171C.
- **Invoice Verification (MIRO):** Automated 3-way match — PO, GR, and invoice — before any payable is created.
- **Supplier Invoices List (MIR5):** Live reporting on invoice status, amounts, and payment due dates.
- **Zero Manual Accounting:** Auto-postings across Stock Account, GR/IR Clearing, and Payables — no manual journal.

3. Tech Stack

Component	Technology	Role in P2P Process
ERP Platform	SAP S/4HANA (Trial)	Core transaction and financial engine
MM Module	SAP Materials Management	Procurement, GR, and inventory control
FI Module	SAP Financial Accounting	Automatic accounting document generation
Database	SAP HANA In-Memory DB	Real-time processing and analytics
UI	SAP Fiori (Browser-based)	Modern UX for all SAP transactions
Reporting	ME2M, MB52, MIR5	Procurement and invoice analytics

4. SAP Screenshots — Step-by-Step Execution

Steps 1 & 2: Create Purchase Order (ME21N) → Confirmed PO 4500064107

Step 1 — Standard PO (Type NB) for CoService Inc (USSU-VSF04), Purch. Org. 1710, Co. Code 1710; CP-RM1 Steel, 100 PC @ USD 500/PC = USD 50,000. Step 2 — SAP generated PO 4500064107, status: In Approval, date 04/20/2026, currency USD.

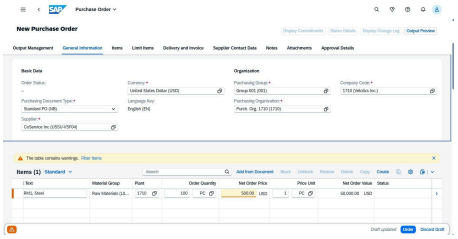

Fig 1: ME21N — New Purchase Order

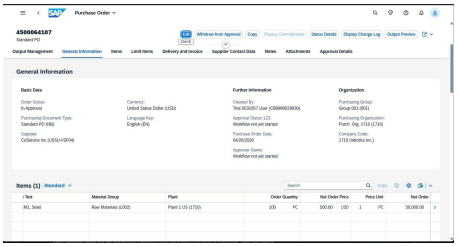

Fig 2: PO 4500064107 Confirmed

Steps 3 & 4: Post Goods Receipt (MIGO) → GR Success

Step 3 — GR posted against PO 4500064107, Movement Type 101; 100 PC CP-RM1 Steel to Storage Loc. 171C, Plant 1710, Unrestricted-Use. Step 4 — Material Document 5000064148/2026 created; auto-posting: Debit Stock / Credit GR/IR, USD 50,000.

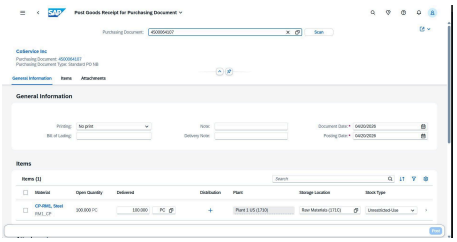

Fig 3: MIGO — Goods Receipt

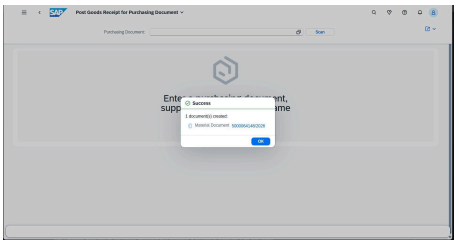

Fig 4: GR Success — Mat. Doc. 5000064148

Step 5: Logistics Invoice Verification (MIRO)

Supplier invoice USD 50,000 entered for USSU-VSF04, Ref INV-2026-001, Co. Code 1710. Reconciliation Account 21100000 (Payables Domestic) auto-populated from vendor master. Invoice & posting date: 04/20/2026.

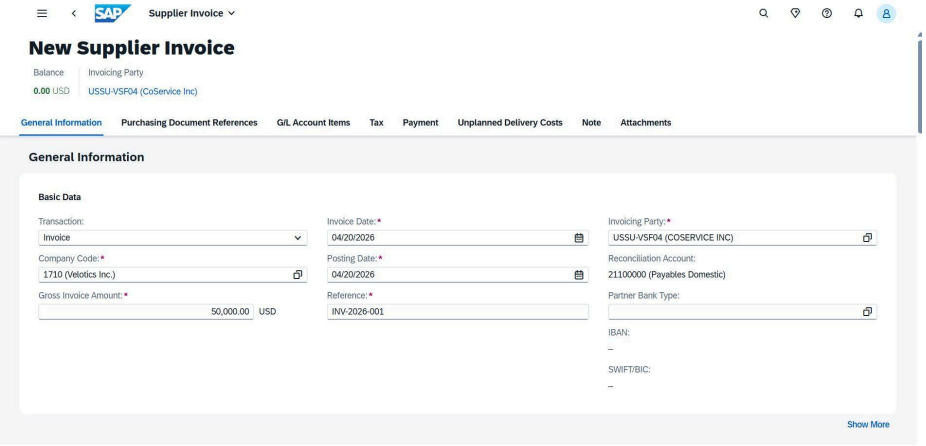

Figure 5: MIRO — Supplier Invoice USD 50,000, Ref INV-2026-001, Recon A/c 21100000

Step 6: 3-Way Match — Purchasing Document Reference

PO 4500064107/10 linked to invoice. SAP validated CP-RM1 Steel — 100 PC, USD 50,000 — completing the three-way match between PO, GR 5000064148/2026, and the invoice.

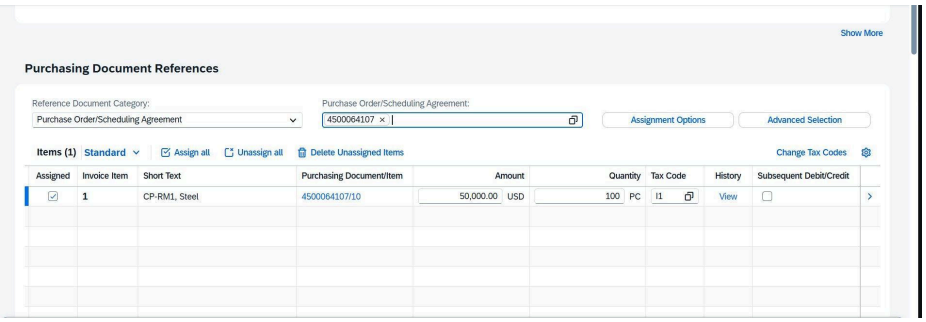

Figure 6: MIRO — PO 4500064107 assigned, 3-Way Match confirmed (100 PC, USD 50,000)

Step 7: Supplier Invoices List — Invoice 5105662821/2026 Posted

Invoice 5105662821/2026: Status: Posted, USD 50,000, Co. Code 1710, date 04/20/2026, Ref INV-2026-001, Entry Type Online, Cash Discount Due 05/04/2026.

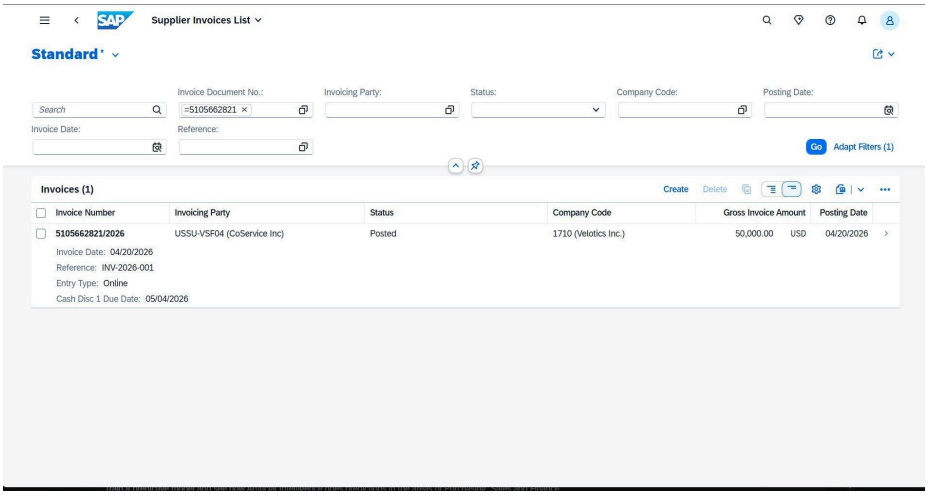

Figure 7: Supplier Invoices List — Invoice 5105662821/2026, Status: Posted

5. Unique Points & Key Learnings

- **Real SAP Execution:** PO 4500064107, GR 5000064148/2026, and Invoice 5105662821/2026 are live documents from SAP S/4HANA trial — not simulated data.
- **Automated 3-Way Match:** SAP verified PO, GR, and Invoice across 100 PC, USD 500/PC, USD 50,000 total — blocking payment for any unverified delivery.
- **Zero Manual Accounting:** GR auto-debited Stock and credited GR/IR; MIRO reversed GR/IR and credited Payables A/c 21100000 — no manual journal entry at any stage.
- **Complete Audit Trail:** Six independently numbered SAP documents form an unbroken chain from PR 10000007 to posted vendor payable 5105662821/2026.
- **Organisational Controls:** Purch. Org. 1710, Group 001, Plant 1710, and Storage Loc. 171C enforce structured spending authority and data ownership.

6. Future Improvements

- **Automatic Payment Run (F110):** Clear the USD 50,000 payable before the 05/04/2026 cash discount due date, completing the full P2P financial cycle.
- **INR Currency & GST:** In a live Indian environment, use INR with IGST/CGST/SGST tax codes in MIRO, replacing the trial-system USD limitation.
- **Request for Quotation (ME41/ME47):** Competitive bidding from multiple vendors before PO creation to ensure optimal pricing and supplier selection.
- **Release Strategy / Approval Workflow:** Value-based PO release strategies requiring authorised manager approval above a defined threshold.
- **SAP Analytics Cloud (SAC):** Connect P2P data to SAC dashboards for real-time spend analysis, supplier scorecards, and AP aging reports.

7. P2P Document Trail — Summary

#	Process	T-Code	Document No.	Outcome
1	Purchase Requisition	ME51N	10000007	INR 3,00,000 | 200 PC
2	Purchase Order	ME21N	4500064107	USD 50,000 | 100 PC
3	Goods Receipt	MIGO	5000064148/2026	Stock updated — 171C
4	Invoice Verification	MIRO	5105662821/2026	USD 50,000 | Posted
5	3-Way Match	Auto	System-verified	PO = GR = Invoice ✓

— End of Project Documentation —

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